

Bay Area Real Estate Prediction Model & Dashboard

Team 211 · CSE 6242
Cody Bradshaw, Divya Raj, Krishnaprasath Ramakrishnasubramanian, Linlin Guo
Georgia Institute of Technology

INTRODUCTION / MOTIVATION

The Problem/Introduction

Buying a home in the Bay Area is one of the most complex financial decisions a family can make.

Median prices **\$1.15M – \$1.50M** across Santa Clara, San Mateo, and Alameda counties

Price variation **>40%** across zip codes within the same county

Buyers must juggle property attributes, neighborhood quality, affordability constraints, and market timing which are currently scattered across Zillow, Redfin, and city databases with **no unified decision-support tool**. We try to solve this unmet need with our interactive dashboard product.

MACHINE LEARNING APPROACH

5 regression models trained on 80/20 split across around 50,000 Bay Area property records. Our prediction function accepts a street address, geocodes it, engineers the relevant features identically to training, and returns a **Gradient Boosting**-predicted price.

Model	R ²	MAPE	RMSE
Gradient Boosting ★	0.927	11.3%	\$239K
XG Boost	0.927	11.4%	\$241K
Random Forest	0.922	11.6%	\$248K
Ridge Regression	0.899	15.9%	\$283K
Linear Regression	0.899	15.9%	\$283K

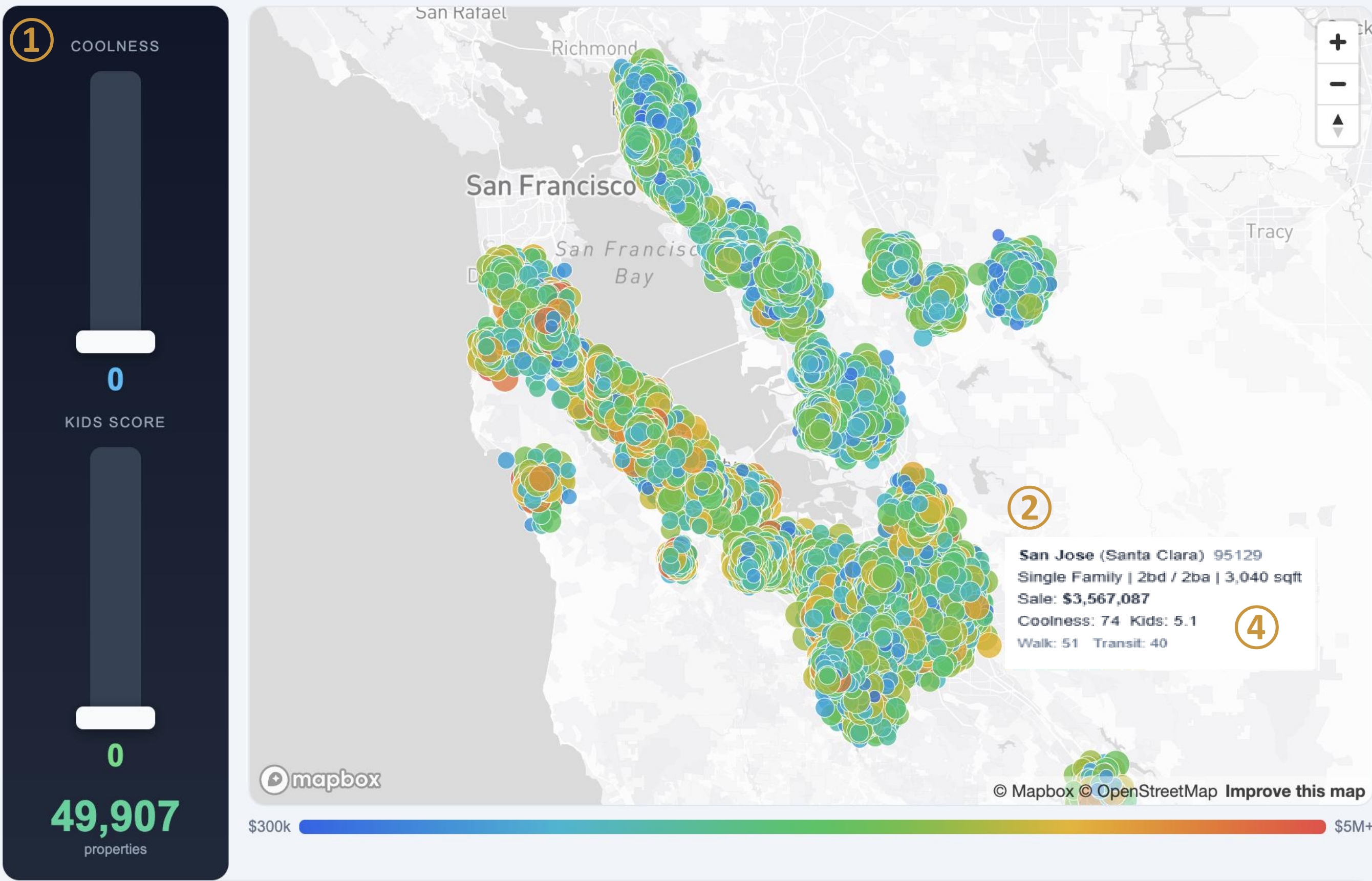
Table: Different ML model performance

INTERACTIVE D3.JS DASHBOARD

Dual-layered visualization transitioning from regional affordability trends to individual property analysis:

- Coolness & Kid Friendly** filterability for neighborhood needs
- Interactive tooltips** with Kid-Friendly & Cool score bar charts on hover
- Affordability Calculator** providing cost realistic filtering with income & downpayment chosen
- Property-level predictions** via Gradient Boosting backend for any Bay Area address

Explore by Coolness & Kids Score



Affordability Calculator

Enter your gross monthly income. We use **60%** for housing, **20%** down payment, and today's 30-year fixed rate.

GROSS MONTHLY INCOME (\$)

12000

30-YR MORTGAGE RATE (%)

6.11

DOWN PAYMENT (%)

20

HOUSING BUDGET (% OF INCOME)

60

Monthly Budget

\$7,200

Max Home Price

\$1,483,580

Down Payment

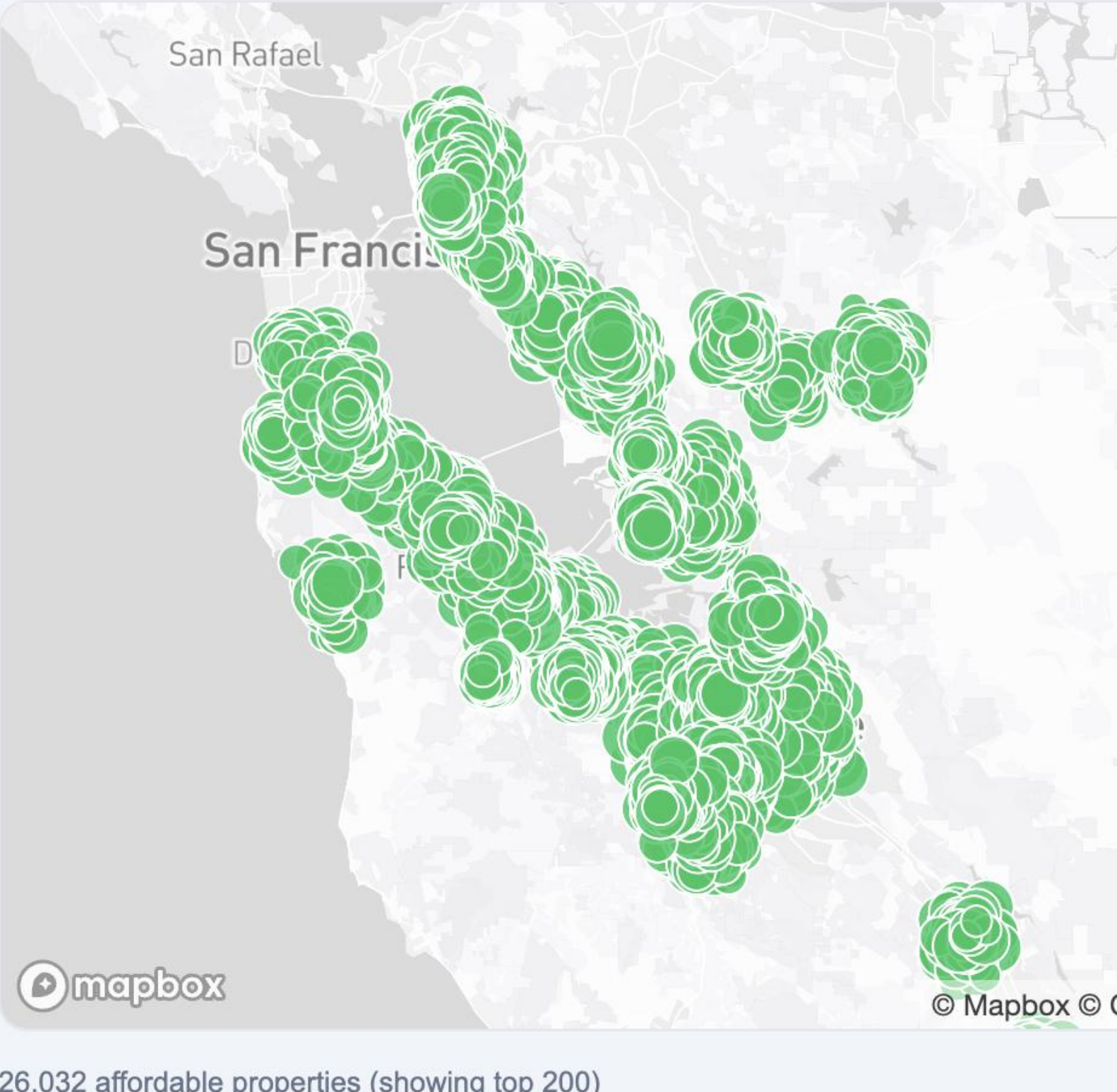
\$296,716

Loan Amount

\$1,186,864

Monthly P&I

\$7,200



SUMMARY & CONCLUSIONS

This project proves that housing search can be improved through an integrated, decision-oriented framework that combines price prediction, affordability analysis, and neighborhood quality assessment within a single interactive dashboard. The dashboard translates these modeling results into a practical visual tool, allowing users to explore Bay Area properties by budget, Coolness, and Kid-Friendly score across 143 zip codes while comparing homes in a more personalized and interpretable manner.

Using approximately 50,000 property records, the model achieved strong predictive performance, and the proposed neighborhood scores contributed additional explanatory value beyond traditional listing platforms. Together, these components illustrate how data-driven, user-centered design can support more informed housing decisions in competitive real estate markets.

DATA

5 sources · 3 Bay Area counties · 143 zip codes

Zillow Research
Property-level sale prices

OpenStreetMap (OSMnx)
park polygons

WalkScore API
Walk/Bike/Transit scores

Redfin Market Tracker
Aggregate market stats

TripAdvisor
Restaurants & attractions

Data Sourced by Scraping / Downloading

- Around 50,000 properties · 143 zip codes ·